Chapter 2. Simple Random Sampling

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1 Introduction

Definition 2.1

Simple random sampling (SRS) is the procedure in which every possible sample of size n has the same chance of being selected; in particular every population unit has the same chance of being in the sample. Without replacement each of the $\binom{N}{n}$ samples has probability $1/\binom{N}{n}$.

A simple random sample can be drawn by the lottery method or with random number tables / computer programs (`sample(N, n)` in R).

Setting. The population values are $u_1, \dots, u_N$ with

$$\mu = \frac{1}{N}\sum_{i=1}^{N} u_i, \qquad \sigma^2 = \frac{1}{N}\sum_{i=1}^{N}(u_i-\mu)^2, \qquad S^2 = \frac{1}{N-1}\sum_{i=1}^{N}(u_i-\mu)^2,$$

and $y_1, \dots, y_n$ is an SRS drawn without replacement (WOR). The **sampling fraction** is $f = n/N$ and $1 - f$ is the **finite population correction** (fpc).

2 Estimation of the Population Mean

Motivation. In survey sampling the population is a fixed finite list and the only randomness comes from which units we select. Under simple random sampling the sample mean is unbiased, and its variance carries the finite population correction $(N - n)/N$, which vanishes when we sample everyone.

Theorem 2.2: Sample mean under SRS (WOR)

Let $\bar{y} = \frac{1}{n}\sum_{i=1}^{n} y_i$ and $s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(y_i - \bar{y})^2$. Then

(i) $\mathrm{E}(\bar{y}) = \mu$;

(ii) $\mathrm{Var}(\bar{y}) = \dfrac{N-n}{N-1}\dfrac{\sigma^2}{n} = \dfrac{N-n}{N}\dfrac{S^2}{n} = \dfrac{S^2}{n}(1-f)$;

(iii) $\mathrm{E}(s^2) = S^2$, hence $\widehat{\mathrm{Var}}(\bar{y}) = \dfrac{s^2}{n}(1-f)$ is unbiased for $\mathrm{Var}(\bar{y})$.

Derivation: proof of the theorem

Marginal distribution of one draw. By symmetry each y_i equals u_k with probability $1/N$, so

$$\mathrm{E}(y_i) = \sum_{k=1}^{N} u_k \frac{1}{N} = \mu, \qquad \mathrm{Var}(y_i) = \sum_{k=1}^{N}(u_k - \mu)^2 \frac{1}{N} = \sigma^2 \quad \text{for all } i.$$

(i) follows: $\mathrm{E}(\bar{y}) = \frac{1}{n}\sum \mathrm{E}(y_i) = \mu$.

Covariance of two draws. For $i \neq j$ the pair (y_i, y_j) is (u_k, u_l), $k \neq l$, with probability $\frac{1}{N(N-1)}$. Hence

$$\begin{aligned}
\mathrm{Cov}(y_i, y_j) &= \mathrm{E}(y_i y_j) - \mu^2 = \frac{1}{N(N-1)}\sum_{k\neq l} u_k u_l - \frac{1}{N^2}\Big(\sum_k u_k\Big)^2 \\
&= \frac{1}{N(N-1)}\Big[\Big(\sum_k u_k\Big)^2 - \sum_k u_k^2\Big] - \frac{1}{N^2}\Big(\sum_k u_k\Big)^2 \\
&= \frac{1}{N(N-1)}\Big[\frac{1}{N}\Big(\sum_k u_k\Big)^2 - \sum_k u_k^2\Big] = -\frac{1}{N(N-1)}\Big(\sum_k u_k^2 - N\mu^2\Big) \\
&= -\frac{1}{N(N-1)}\sum_k (u_k - \mu)^2 = -\frac{\sigma^2}{N-1}.
\end{aligned}$$

Variance of the mean.

$$\begin{aligned}
\mathrm{Var}(\bar{y}) &= \frac{1}{n^2}\Big[\sum_i \mathrm{Var}(y_i) + 2\sum_{i<j}\mathrm{Cov}(y_i, y_j)\Big] = \frac{1}{n^2}\Big[n\sigma^2 - n(n-1)\frac{\sigma^2}{N-1}\Big] \\
&= \frac{\sigma^2}{n}\Big(1 - \frac{n-1}{N-1}\Big) = \frac{\sigma^2}{n}\frac{N-n}{N-1} = \frac{S^2}{n}\frac{N-n}{N},
\end{aligned}$$

using $N\sigma^2 = (N-1)S^2$.

Unbiasedness of s^2. Using $\sum(y_i - \bar{y})^2 = \sum(y_i - \mu)^2 - n(\bar{y} - \mu)^2$,

$$\mathrm{E}(s^2) = \frac{1}{n-1}\big[n\sigma^2 - n\mathrm{Var}(\bar{y})\big] = \frac{\sigma^2}{n-1}\Big(n - \frac{N-n}{N-1}\Big) = \frac{\sigma^2}{n-1}\cdot\frac{N(n-1)}{N-1} = \frac{N}{N-1}\sigma^2 = S^2.$$

Therefore $\mathrm{E}[\widehat{\mathrm{Var}}(\bar{y})] = \frac{N-n}{Nn}\mathrm{E}(s^2) = \mathrm{Var}(\bar{y})$.

Remark 2.3: SRS with replacement

If the draws are made with replacement, the y_i are iid with mean μ and variance σ^2. Then $\mathrm{Cov}(y_i, y_j) = 0$ (e.g. for 0/1 values $\mathrm{E}(y_i y_j) = P(y_i = 1)P(y_j = 1) = \frac{N_1}{N}\frac{N_1}{N}$), so

$$\mathrm{Var}(\bar{y}) = \frac{\sigma^2}{n}, \qquad \mathrm{E}(s^2) = \frac{1}{n-1}\Big(n\sigma^2 - n\frac{\sigma^2}{n}\Big) = \sigma^2, \qquad \widehat{\mathrm{Var}}(\bar{y}) = \frac{s^2}{n}.$$

Without replacement the variance is smaller by the factor $\frac{N-n}{N-1}$; the fpc matters only when f is not small.

Confidence interval and bound. A $100(1-\alpha)\%$ confidence interval for μ is

$$\bar{y} \pm z_{\alpha/2}\sqrt{1-f}\,\frac{s}{\sqrt{n}},$$

and with $\alpha = 0.05$ we use the 2-SD bound $B = 2\sqrt{\widehat{\mathrm{Var}}(\bar{y})} = 2\sqrt{\frac{s^2}{n}(1-f)}$.

Example 2.4: Tractors per village

Estimate the average number of tractors per village; $N = 69$, $n = 10$, $\bar{y} = 16.7$, $s^2 = 81.344$.

$$\widehat{\mathrm{Var}}(\bar{y}) = \frac{s^2}{n}\cdot\frac{N-n}{N} = \frac{81.344}{10}\cdot\frac{59}{69} = 6.956, \qquad 2\sqrt{6.956} = 5.27,$$

so the interval is $16.7 \pm 5.27 = [11.43,\ 21.97]$.

3 Estimation of the Population Total

Motivation. Often the quantity of interest is a total (total income, total harvest). Scaling the mean by N gives an unbiased estimator whose variance is N^2 times that of the mean.

The population total is $\tau = N\mu$, and with N known the natural estimator is $\hat{\tau} = N\bar{y}$.

Theorem 2.5

$\mathrm{E}(\hat{\tau}) = \tau$ and

$$\mathrm{Var}(\hat{\tau}) = N^2\frac{N-n}{N-1}\frac{\sigma^2}{n} = N^2\frac{N-n}{N}\frac{S^2}{n}, \qquad \widehat{\mathrm{Var}}(\hat{\tau}) = N^2\frac{N-n}{N}\frac{s^2}{n} = N^2\frac{s^2}{n}(1-f).$$

A $100(1-\alpha)\%$ interval is $\hat{\tau} \pm z_{\alpha/2}\sqrt{\widehat{\mathrm{Var}}(\hat{\tau})}$, typically $\hat{\tau} \pm 2\sqrt{\widehat{\mathrm{Var}}(\hat{\tau})}$.

Example 2.6: Total number of tractors

$\hat{\tau} = 69 \times 16.7 = 1152.3$, $\widehat{\mathrm{Var}}(\hat{\tau}) = 69^2 \times 6.956 = 33{,}117$, and $2\sqrt{33117} = 364.0$. The interval is $1152.3 \pm 364.0 = [788,\ 1516]$.

4 Sample Size for Estimating μ and τ

To estimate μ with a bound B, set $2\sqrt{\mathrm{Var}(\bar{y})} = B$:

$$B = 2\sqrt{\frac{N-n}{N-1}\frac{\sigma^2}{n}} \implies n = \frac{N\sigma^2}{(N-1)D+\sigma^2} = \frac{NS^2}{ND+S^2}, \qquad D = \frac{B^2}{4}.$$

Derivation

Write $D = B^2/4$. Then $D\,n(N-1) = (N-n)\sigma^2$, i.e. $n[(N-1)D+\sigma^2] = N\sigma^2$. The second form follows from $\sigma^2 = \frac{N-1}{N}S^2$.

For the total, $B = 2\sqrt{N^2\frac{N-n}{N}\frac{S^2}{n}}$ gives

$$n = \frac{NS^2}{ND+S^2}, \qquad D = \frac{B^2}{4N^2}.$$

In practice S^2 (or σ^2) is unknown and replaced by a prior estimate or a pilot-sample s^2.

Example 2.7: Tractors, continued

With $s^2 = 81.344$ and $B = 3$ ($D = 2.25$),

$$n = \frac{69 \times 81.344}{69 \times 2.25 + 81.344} = \frac{5612.7}{236.6} = 23.72 \;\Rightarrow\; n = 24,$$

so 14 more villages are needed beyond the 10 already sampled.

5 Estimation of a Population Proportion

Let $y_i = 1$ if the ith unit has the characteristic and 0 otherwise. Then the population proportion is the population mean of the 0/1 values, $p = \mu$, and

$$\hat{p} = \bar{y} = \frac{1}{n}\sum_{i=1}^{n} y_i, \qquad \hat{q} = 1 - \hat{p}.$$

For 0/1 values $\sigma^2 = pq$ and $S^2 = \frac{N}{N-1}pq$.

Theorem 2.8: Proportion under SRS

$$\mathrm{E}(\hat{p}) = p, \qquad \mathrm{Var}(\hat{p}) = \frac{pq}{n}\frac{N-n}{N-1}, \qquad \widehat{\mathrm{Var}}(\hat{p}) = \frac{\hat{p}\hat{q}}{n-1}\frac{N-n}{N} = \frac{\hat{p}\hat{q}}{n-1}(1-f),$$

and $\mathrm{E}[\widehat{\mathrm{Var}}(\hat{p})] = \mathrm{Var}(\hat{p})$.

Derivation: why the divisor is $n-1$

$$\mathrm{E}(\hat{p}\hat{q}) = \mathrm{E}(\hat{p}) - \mathrm{E}(\hat{p}^2) = p - [\mathrm{Var}(\hat{p}) + p^2] = pq - \frac{pq}{n}\frac{N-n}{N-1}$$
$$= pq\,\frac{n(N-1)-(N-n)}{n(N-1)} = \frac{N(n-1)}{(N-1)n}\,pq.$$

Hence

$$\mathrm{E}\Big[\frac{\hat{p}\hat{q}}{n-1}\frac{N-n}{N}\Big] = \frac{N-n}{N(n-1)}\cdot\frac{N(n-1)}{(N-1)n}pq = \frac{pq}{n}\frac{N-n}{N-1} = \mathrm{Var}(\hat{p}).$$

By contrast the plug-in $\frac{\hat{p}\hat{q}}{n}(1-f)$ is biased downward: $\mathrm{E}\big[\frac{\hat{p}\hat{q}}{n}(1-f)\big] \neq \mathrm{Var}(\hat{p})$. This is the same phenomenon as s^2 versus the divisor-n variance, since $s^2 = \frac{n}{n-1}\hat{p}\hat{q}$ for 0/1 data.

Example 2.9: Student vote

An SRS of $n = 120$ from $N = 2400$ students; 72 support candidate A. Then $\hat{p} = 0.6$ and

$$\widehat{\mathrm{Var}}(\hat{p}) = \Big(1 - \frac{120}{2400}\Big)\frac{0.6\times 0.4}{119} = 0.001916, \qquad 2\sqrt{0.001916} = 0.0875,$$

so the interval is $0.6 \pm 0.0875 = (0.512,\ 0.688)$.

6 Sample Size for Estimating p

Replace σ^2 by pq in the formula for μ:

$$n = \frac{Npq}{(N-1)D + pq}, \qquad D = \frac{B^2}{4} \quad (\text{or } D = B^2/1.96^2 \text{ with the exact normal quantile}).$$

With no prior information take $p = 1/2$, which maximizes pq and gives a conservative n.

Example 2.10

For $N = 2400$ and $B = 0.05$: $D = 0.05^2/4 = 0.000625$ and

$$n = \frac{2400\times 0.25}{2399\times 0.000625 + 0.25} = \frac{600}{1.7494} = 342.98 \ \Rightarrow\ n = 343.$$

7 Towards Stratified Sampling

Consider estimating the total first-hospitalization cost of kidney-stone patients. Two regions with very different incidence rates are surveyed separately:

	Carolinas	Rockies
n	363	258
$\bar{y}$	1350	1150

What is the total cost? Pooling all 621 observations into one mean ignores the fact that the regions were sampled separately and have different population sizes. The answer is to weight each regional mean by its population size, which is the stratified estimator of Chapter 3.